A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy & material cycles: natural vs human systems - Set 1

Mark scheme

AQA spec 3.6.2-3.6.3; Sustainability; textbook pp. 373-374

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis – human processes usually use high energy-density non-renewable fossil fuels **1** Natural processes occur at low temperatures because enzymes lower the activation energy of reactions – human processes need high temperatures and pressures **1** Natural processes produce no toxic wastes / their products are raw materials for other processes – industrial processes emit CO2 and NOx **1** So natural fixation is sustainable indefinitely with a low carbon footprint **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hydrological cycle (evaporation, precipitation) **1** Carbon cycle / photosynthesis **1** Atmospheric circulation / winds **1** Thermohaline circulation (density differences from heating / evaporation) **1** Decomposition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting **1** Anaerobic digestion and composting treat wastes – enzymes in industry (biological washing powders, biofuels from algae / microorganisms) **1** Heat pumps and solar thermal capture low-density heat – low-temperature manufacturing using biological polymers – fuel cells instead of combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates **1** Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials **1 Human systems are linear:** extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult **1** Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life **1** They leach from landfill or are released by incineration – persistent, bioaccumulating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **incineration** → burning waste, usually with energy recovery • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) **1** Easy identification of materials (coding) and separation of components so they can be recycled **1** Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) **1** Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
03.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Hydrological cycle **1** Carbon cycle **1** Nitrogen cycle **1** Atmospheric circulation **1** Thermohaline circulation **1** Photosynthesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Enzymes reduce the activation energy of reactions **1** So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
03.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Haber process (nitrogen fixation) **1** Incineration of wastes (decomposition) **1** Many manufacturing processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **Incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Energy:** low energy-density renewable (solar) / non-renewable fossil fuels **1 Materials:** cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion **1 Wastes:** biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate **1 Temperature:** low / high (Haber process, incineration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time • **incineration** → burning waste, usually with energy recovery **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) **1** They are re-used in cycles so there is no depletion of raw materials or build-up of wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
04.3	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** Use often disperses the minerals (eg in products spread over the world / corrosion) **OR** produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** They cannot easily be removed from end-of-life appliances and are not biodegradable **1** Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	

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